

CN Unit 1 Question Bank — Introduction & Physical Layer

Computer Networks (CN) · Unit 1 · Introduction & Physical Layer · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Compare OSI Reference Model with Multiplexing. When would you prefer one over the other?
2. Compare TCP/IP Model with OSI Reference Model. When would you prefer one over the other?
3. What is meant by Network Topologies? Discuss its advantages and limitations.
4. Discuss the role of Transmission Media in Introduction & Physical Layer.
5. Explain Multiplexing with a neat diagram and a suitable example.
6. Discuss the role of Switching Techniques in Introduction & Physical Layer.

Part B — Long answer (10 marks)

1. Explain OSI Reference Model with a neat diagram and a suitable example.
2. Explain TCP/IP Model with a neat diagram and a suitable example.
3. Illustrate Network Topologies with a worked example.
4. Explain Transmission Media with a neat diagram and a suitable example.
5. Illustrate Multiplexing with a worked example.
6. Write short notes on Switching Techniques.
7. Compare OSI Reference Model with TCP/IP Model. When would you prefer one over the other?
8. What is meant by TCP/IP Model? Discuss its advantages and limitations.
9. What is meant by Network Topologies? Discuss its advantages and limitations.
10. Illustrate Transmission Media with a worked example.
11. Write short notes on Multiplexing.
12. What is meant by Switching Techniques? Discuss its advantages and limitations.